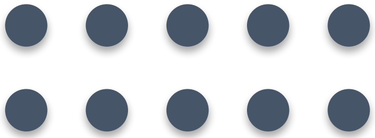
01 Sets (Set Concepts)

OPENING VISUAL

How many stones are in my hand?

FINITE



Think alone

INFINITE

Pair talk

Today's mission

Goal: I can distinguish between finite and infinite sets using

1 Learn

Use: Can We Count Everything?, Defining Our Sets

2 Practise

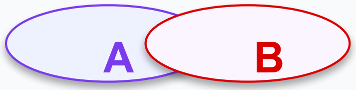
Show: answer with one reason or example.

3 Show

Routine: listen, talk, try, explain.

4 Reflect

Explain in your own words.



Starter: think first

1

THINK

How many stones are in my hand?

30 seconds

2

PAIR

compare your first answer with a partner.

with partner

3

SHARE

connect your answer to Can We Count Everything?.

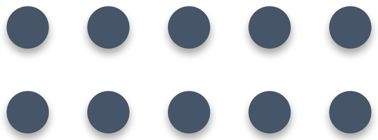
whole class

Can We Count Everything?

CONTENT MAP

Concrete exploration of countable objects

FINITE



Practical counting of stones

INFINITE

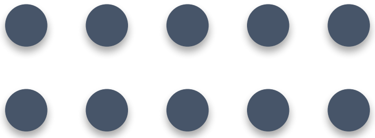
Attempting to count grains of sand

Defining Our Sets

MODEL MAP

Worked example

FINITE



Finite Set: Has a countable number of members

INFINITE

Infinite Set: Has members that go on forever

Try it together

PRACTICE FLOW

Answer with evidence

? Can you tell me exactly how many grains are in that pile of sand?

answer + reason

? Which count was easier to finish?

answer + reason

? What is a finite set?

answer + reason

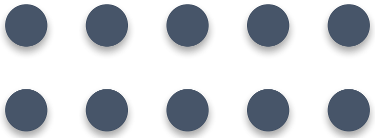
Use evidence from the lesson, not guessing.

Activity challenge

GROUP INFOGRAPHIC

Challenge

FINITE



In groups of five, learners count a pile of 10

INFINITE

Write the definitions of finite and infinite

Share-out: explain one result, then improve it.

Exit ticket

EXIT TICKET

Answer: Can a very large set still be finite? (Yes, if it has an end).

Use one complete sentence.

Use one key word: Set

Hand in or read your answer aloud.